Installing the Jetpack Environment

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Some users may need to use the NVIDIA built-in system component environment. After installing a clean system and successfully entering the system desktop, you can follow the tutorial below to install the component environment! If you are using a factory image, the Jetpack environment is already installed and does not need to be installed again.

```
The entire process requires a network connection. Some components may require a
VPN. If you cannot complete the component installation, please use the factory
image.
```

1. Installing the Jetpack Environment

Change the environment source

```
sudo apt update
```

Install the Jetpack environment. Type 'y' and press Enter to confirm.

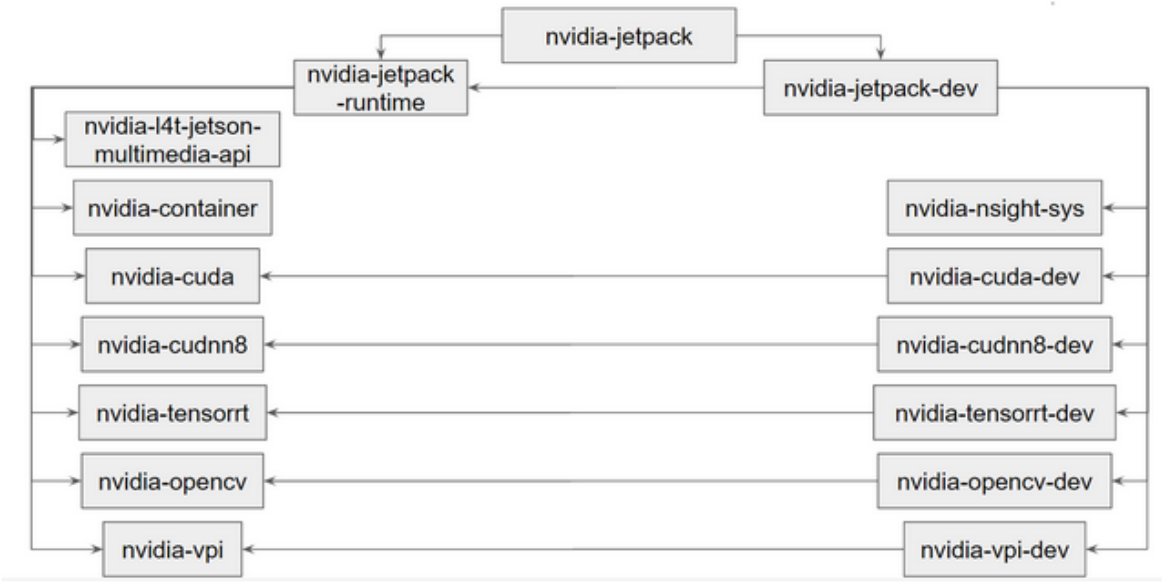
```
sudo apt install nvidia-jetpack
```

```
jetson@ubuntu:~$ sudo apt install nvidia-jetpack
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  vim vim-common vim-runtime
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  cuda-cccl-13-0 cuda-command-line-tools-13-0 cuda-compiler-13-0 cuda-crt-13-0
  cuda-cudart-13-0 cuda-cudart-dev-13-0 cuda-culibos-dev-13-0
  cuda-cuobjdump-13-0 cuda-cupti-13-0 cuda-cupti-dev-13-0 cuda-cuxxfilt-13-0
  cuda-documentation-13-0 cuda-driver-dev-13-0 cuda-gdb-13-0
  cuda-libraries-13-0 cuda-libraries-dev-13-0 cuda-nsight-compute-13-0
  cuda-nsight-systems-13-0 cuda-nvcc-13-0 cuda-nvdisasm-13-0
  cuda-nvml-dev-13-0 cuda-nvprune-13-0 cuda-nvrtc-13-0 cuda-nvrtc-dev-13-0
```

The download requires approximately 14.1GB of space, and the installation time is relatively long. Please be patient.

```
  python3.12-vpi4 tensorrt tensorrt-libs vpi4-dev vpi4-python-src vpi4-samples
0 upgraded, 124 newly installed, 10 to remove and 328 not upgraded.
Need to get 8,688 MB of archives.
After this operation, 14.1 GB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

NVIDIA-JETPACK is a metapackage that will install the following components:



2. Adding Environment Variables

Open the system terminal and run the following commands to add them to the environment variables:

```
echo "export PATH=/usr/local/cuda/bin:$PATH" >> ~/.bashrc
echo "export LD_LIBRARY_PATH=/usr/local/cuda/lib64:$LD_LIBRARY_PATH" >>
~/.bashrc
source ~/.bashrc
```

3. Environment Verification

> **Jtop Tool**

Install Jtop using the following commands:

```
sudo apt update && sudo apt upgrade
sudo apt install python3-pip -y
sudo pip3 install -U jetson-stats
```

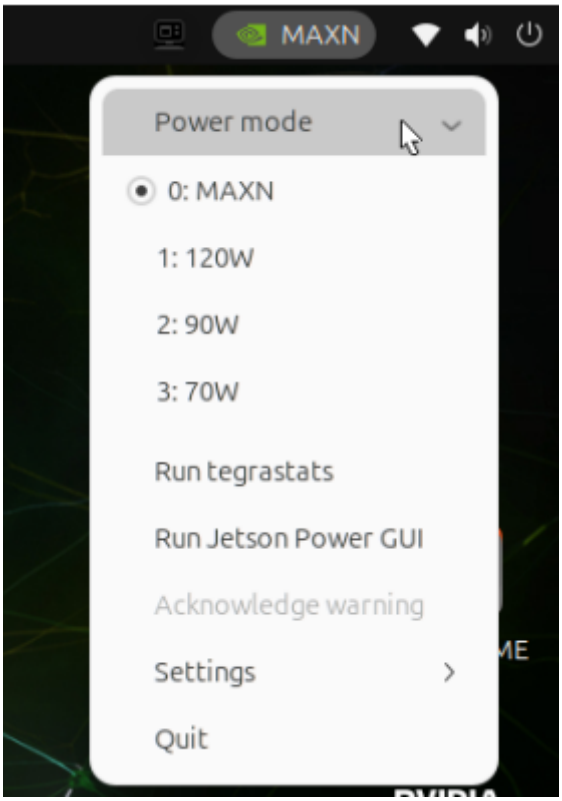
Note: A system restart is required after installation!

> **Optimal Performance Mode**

Enable MAX Power Mode:

```
sudo nvpmodel -m 0
```

Alternatively, locate the NVIDIA icon in the top-right corner of the desktop and change the power mode to MAXN.

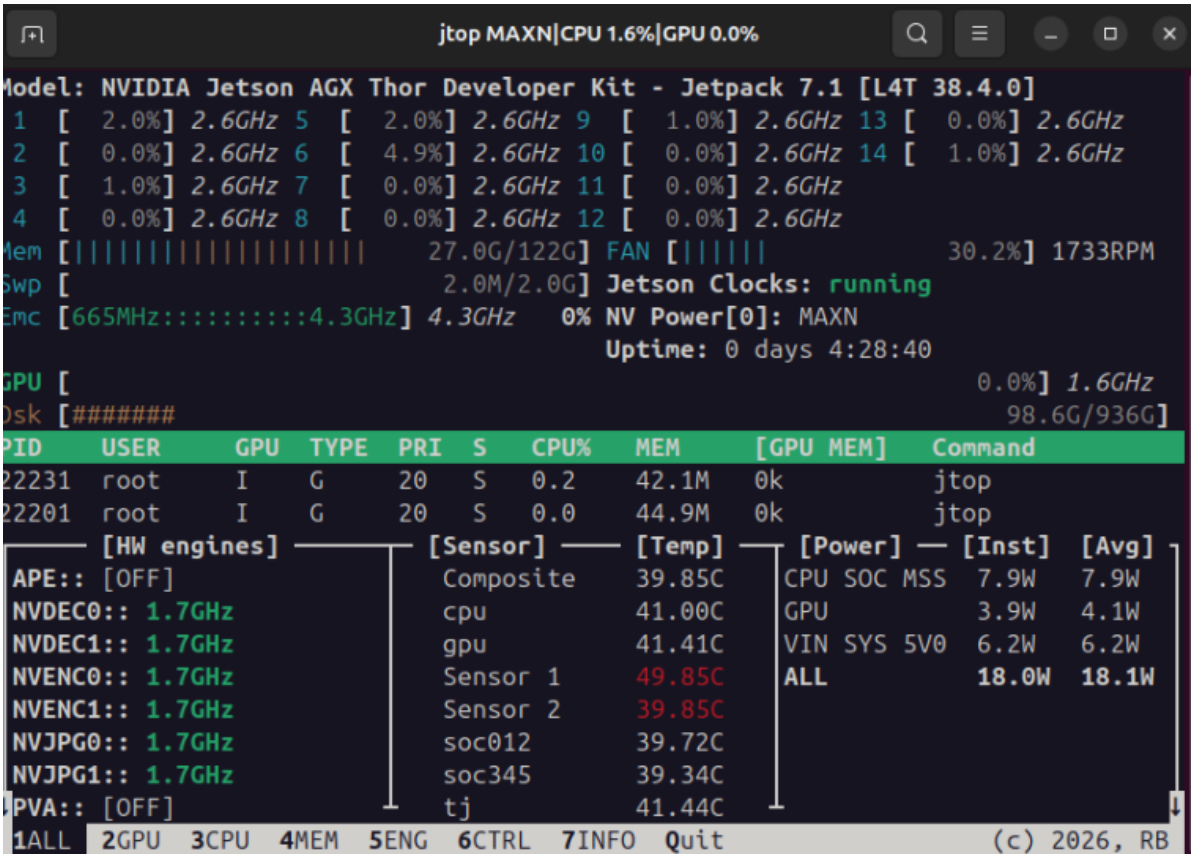


Enable Jetson Clocks: Both the CPU and GPU cores will run at their maximum frequencies.

```
sudo jetson_clocks
```

Use the Jtop tool to view system information:

```
jtop
```



```
jtop 7.1.5 - (c) 2026, Raffaello Bonghi [raffaello@rnext.it]
Website: https://rnext.it/jetson_stats

 Platform                                  Serial Number: [s|XX CLICK TO READ XXX
] Machine: aarch64                         Hardware
  System: Linux                             Model: NVIDIA Jetson AGX Thor Develop
  Distribution: Ubuntu 24.04 Noble Numbat   699-level Part Number: 699-13834-0008
  Release: 6.8.12-tegra                     P-Number: p3834-0008
  Python: 3.12.3                            Module: NVIDIA Jetson AGX Thor (Devel
                                            SoC: tegra264
 Libraries                                  CUDA Arch BIN: 11.0
  CUDA: 13.0.48                             L4T: 38.4.0
  cuDNN: 9.12.0                             Jetpack: 7.1
  TensorRT: 10.13.3.9
  VPI: 4.0.5.0                              Hostname: ubuntu
  Vulkan: 1.4.321                           Interfaces
  OpenCV: 4.8.0 with CUDA: NO               wlP1p1s0: 192.168.12.38
                                            l4tbr0: 192.168.55.1
                                            docker0: 172.17.0.1
```